



Lesson 5.7 — Unit 5 Review and Assessment

Big idea: Unit 5 is roots and rational exponents — different parent shapes, the rules of fractional powers, and the radical-equation trap of extraneous solutions.

Quick review

Even vs. odd roots: even-root domains restrict x ; odd-root domains are all reals.

Transformations: same shift/stretch/reflect rules, anchored at (h, k) .

Rational exponents: $x^{m/n} = \sqrt[n]{x^m}$; usual exponent rules apply.

Radical equations: isolate, raise to the power, ALWAYS check.

Worked example

Worked review. Solve $\sqrt{x+2} = x$. Square: $x+2 = x^2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x-2)(x+1) = 0$. Check $x = 2$: $\sqrt{4} = 2$ ✓. Check $x = -1$: $\sqrt{1} = 1 \neq -1$ ✗. Solution: $x = 2$.

Warm-ups

1. Compute $16^{3/2}$.
2. Domain of $f(x) = \sqrt{x-4}$.
3. Solve $\sqrt{x} = 5$.

Practice

1. Domain and range of $f(x) = \sqrt{x+1} - 3$.
2. Solve $\sqrt{2x+3} = x$.
3. Simplify $x^{1/2} \cdot x^{3/4}$.

Challenge

1. A pendulum's period is $T = 2\pi\sqrt{L/9.8}$ seconds. Solve for L when $T = 2$ s.
2. Solve $x^{2/3} + x^{1/3} - 6 = 0$ (Hint: substitute $u = x^{1/3}$).



Answer key — Lesson 5.7

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $(\sqrt{16})^3 = 64$.
2. $x \geq 4$.
3. $x = 125$.

Practice

1. Domain: $x \geq -1$. Range: $y \geq -3$.
2. Square: $2x + 3 = x^2 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0$. Check $x = 3$: $\sqrt{9} = 3$ ✓. $x = -1$: $\sqrt{1} = 1 \neq -1$ ✗.
Solution: $x = 3$.
3. $x^{5/4}$.

Challenge

1. $T^2 = 4\pi^2 L / 9.8 \Rightarrow L = 9.8 T^2 / (4\pi^2)$. $T = 2$: $L = 39.2 / (4\pi^2) \approx 0.99$ m.
2. Let $u = x^{1/3}$: $u^2 + u - 6 = 0 \Rightarrow (u + 3)(u - 2) = 0 \Rightarrow u = -3$ or 2 . So $x = u^3$: $x = -27$ or 8 .